

EduGuide AI

**AI-Powered Career Guidance Bot for
Mechanical Engineering Students**

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AI- Capstone Project

SELECTED SDG

The Sustainable Development Goal selected for this project is **United Nations SDG 4 – Quality Education**. This SDG focuses on the ensuring inclusive, equitable, and quality education while promoting lifelong learning opportunities for all students.

This SDG was selected since many students, especially in the field of mechanical engineering, struggle to select the right electives and career route due to a lack of proper guidance and understanding. Multiple fields are available in mechanical engineering, including design, manufacturing, robotics, automation, thermal engineering, automotive engineering, and many more. Students frequently struggle to choose the field that best suits their interests and abilities due to the abundance of possibilities.

By providing correct career recommendations, information about various domains, necessary skills, possibilities for further education, and professional opportunities in mechanical engineering, the developed chatbot seeks to deal with this issue. In order to help students

make well-informed decisions about their academic and professional futures, the chatbot serves as an easily available guidance tool.

Therefore, by enabling improved access to educational guidance and assisting students in developing a clearer understanding of their learning and career development, this project directly supports SDG 4.

PROBLEM STATEMENT

In the field of Mechanical Engineering, students often face difficulty in selecting the right career path and electives due to a lack of proper guidance and industry awareness. Mechanical Engineering consists of multiple domains such as Design, Manufacturing, Robotics, Automation, Thermal Engineering, and Automotive Engineering, which creates confusion among students while choosing their area of interest.

Many students are unaware of the skills, software tools, certifications, and technologies required in different domains. They also do not receive personalized guidance based on their interests, strengths, and career goals. As a result, students may select unsuitable electives or career paths, which can negatively affect their learning experience, confidence, and future opportunities.

Undergraduate Mechanical Engineering students who choose their specialization, internships, further education, or career path are mainly affected by this problem. Students could face limited career planning, decreased skill development, a lack of industry readiness, and trouble finding appropriate employment possibilities if this issue is not resolved.

Therefore, there is a need for an intelligent and accessible guidance system that can help students understand various Mechanical Engineering domains and make informed academic and career decisions.

PROPOSED SOLUTION

The proposed solution is an AI-powered chatbot named “EduGuide AI” developed using the Botpress platform. The chatbot helps Mechanical Engineering students identify suitable electives, career opportunities, technical skills, and project ideas based on their interests and academic profile.

The chatbot first collects user information such as:

Name

Current year of study

Career goals

Coding interest

Preferred work style

Technical interests

After collecting this information, the chatbot uses a Profile Analyzer module to evaluate the student's preferences and generate domain-specific scores for areas such as:

- Thermal and Fluid Sciences
- Design and Robotics
- Manufacturing and Automation

Based on the analysis, the AI recommendation engine provides:

- Personalized elective recommendations
- Suggested software tools
- Mini project ideas
- Internship domains
- Career opportunities
- Future industry trends
- Learning roadmap

The chatbot also adapts its responses according to the student's current academic year, ensuring personalized and relevant guidance.

PROJECT FEATURES

FEATURE 1- Personalized User Profiling

Collects:

- year
- interests
- career goals
- coding level

to generate customized recommendations.

FEATURE 2 – AI Career recommendation Engine

Analyzes user profile and suggests:

- suitable domains
- electives
- career opportunities

FEATURE 3 – Skill Roadmap Generator

Provides:

- software recommendations
- mini projects
- technical learning path

FEATURE 4 – Domain Comparison

Helps students compare:

- Robotics
- Thermal
- Manufacturing

based on future scope and required skills.

FEATURE 5 – Industry Trend Awareness

Suggests:

EV industry opportunities

Industry 4.0

Automation

AI in Mechanical Engineering

TECHNOLOGY STACK

The development of this project involved multiple modern technologies and AI-based tools that work together to provide an intelligent and personalized guidance system for Mechanical Engineering students. Each technology plays an important role in ensuring smooth interaction, accurate recommendations, and efficient workflow management within the chatbot system.

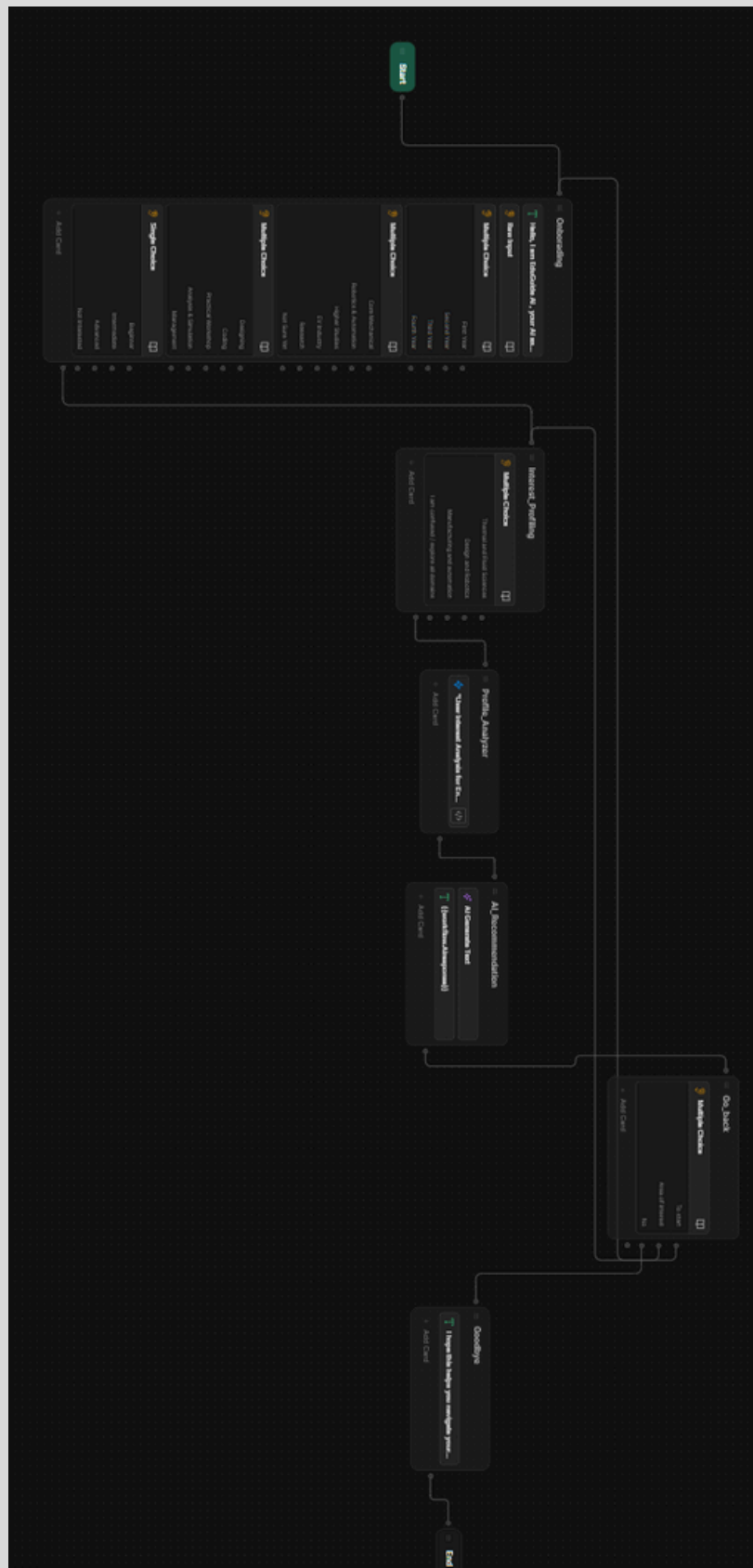
- **Botpress** – Botpress was used as the primary chatbot development platform for designing, managing, and deploying the AI-powered guidance system. It provides a visual workflow builder that helps in creating conversational flows, onboarding processes, and recommendation modules. Botpress also supports integration with AI services and allows easy customization of chatbot behavior according to user requirements.

- **Generative AI** – Generative AI technology was used to provide intelligent and personalized responses to students based on their interests, academic year, and career goals. It helps the chatbot generate dynamic recommendations related to electives, career domains, software tools, internships, and future opportunities in Mechanical Engineering. This makes the interaction more human-like and adaptive.
- **Natural Language Processing (NLP)** – NLP techniques were used to understand and process user inputs effectively. The chatbot can interpret student queries, identify interests, and respond appropriately even when users provide inputs in different styles or formats. NLP improves communication quality and enhances the overall user experience by enabling more natural conversations.
- **Knowledge Base System** – A structured knowledge base was created to store information related to Mechanical Engineering domains, electives, software tools, career opportunities, industry trends, and project ideas. This knowledge repository helps the chatbot provide accurate and domain-specific guidance to students while ensuring consistency in recommendations.
- **Workflow Automation** – Workflow automation was implemented to manage the complete interaction process efficiently.

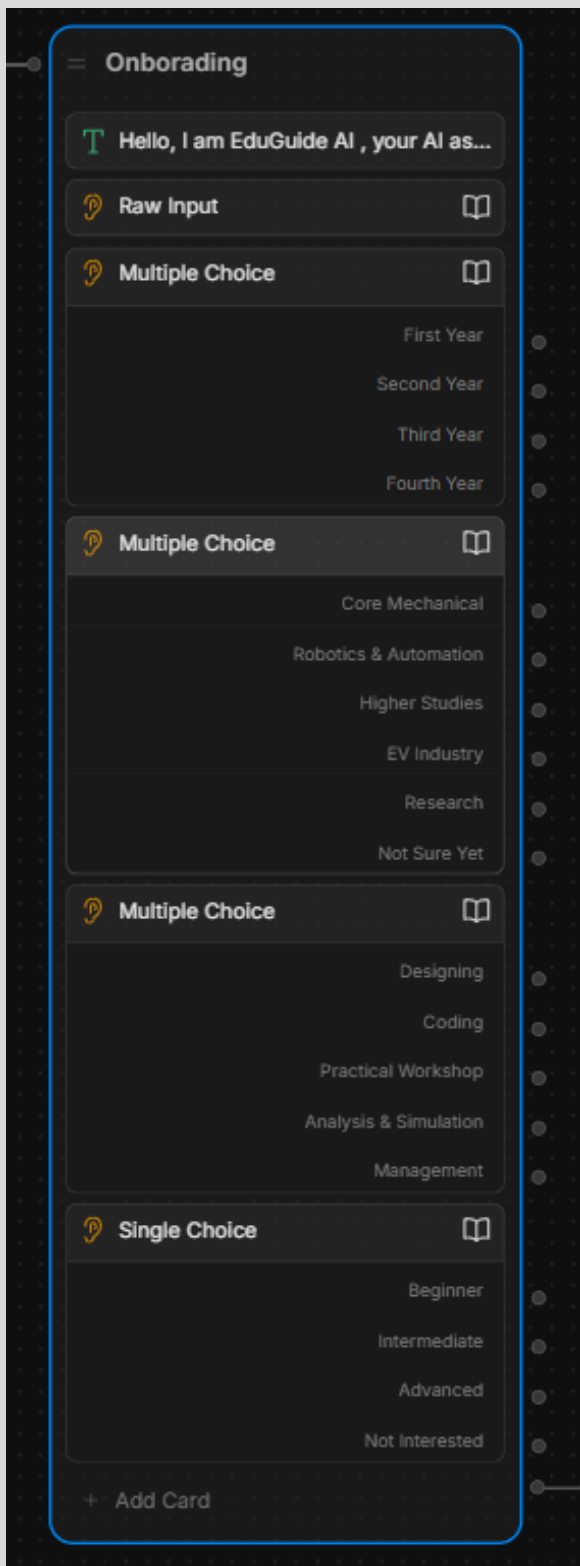
Automated workflows handle onboarding, interest profiling, profile analysis, recommendation generation, and personalized guidance delivery. This ensures that every student receives a smooth and organized experience without manual intervention.

- **AI Recommendation Engine** – An AI-based recommendation mechanism was integrated into the chatbot to analyze student preferences and generate suitable career suggestions. The recommendation engine evaluates factors such as technical interests, coding skills, preferred work style, and academic level to suggest the most relevant Mechanical Engineering domains and electives.
- **Cloud-Based Deployment Support** – The chatbot platform supports cloud-based deployment, allowing the system to be accessed online from different devices. This improves accessibility and ensures that students can interact with the chatbot anytime and from anywhere.
- **User Profiling and Data Handling** – The project also includes user profiling mechanisms that collect and organize student information securely. This data is used to personalize recommendations and improve the relevance of guidance provided by the chatbot.

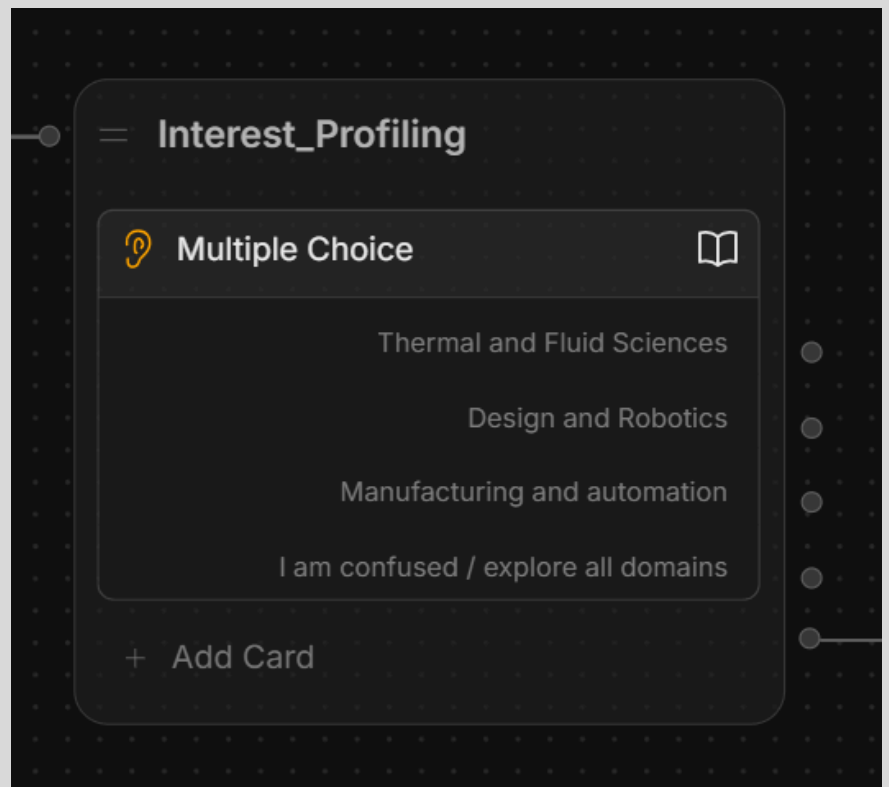
PROJECT SCREENSHOTS



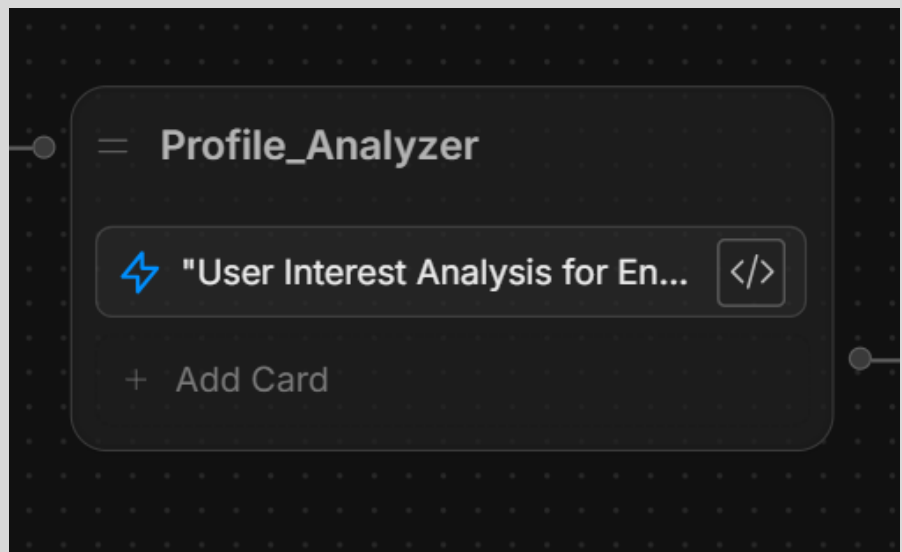
Workflow



Onboarding Node



Interest Profiling node



Profile Analyzer node

</> Execute Code

"User Interest Analysis for Career Path Recommendations"

🔗 Generate your action with AI

Ask Botpress AI to edit code



```
1 // Initial scores
2 let thermal = 0;
3 let robotics = 0;
4 let manufacturing = 0;
5
6 // Interest Based Analysis
7 if(user.interests.includes("Thermal and Fluid Sciences")){
8   thermal += 5;
9 }
10
11 if(user.interests.includes("Design and Robotics")){
12   robotics += 5;
13 }
14
15 if(user.interests.includes("Manufacturing and Automation")){
16   manufacturing += 5;
17 }
18
19 // Coding Interest Analysis
20 if(user.coding === "Advanced"){
21   robotics += 3;
22 }
23
24 if(user.coding === "Intermediate"){
25   robotics += 2;
26 }
27
28 if(user.coding === "Not Interested"){
```

</> Execute Code

"User Interest Analysis for Career Path Recommendations"

🔗 Generate your action with AI

Ask Botpress AI to edit code



```
25   robotics += 2;
26 }
27
28 if(user.coding === "Not Interested"){
29   thermal += 1;
30   manufacturing += 2;
31 }
32
33 // Work Style Analysis
34 if(user.workstyle === "Designing"){
35   robotics += 2;
36 }
37
38 if(user.workstyle === "Practical Workshop"){
39   manufacturing += 3;
40 }
41
42 if(user.workstyle === "Analysis & Simulation"){
43   thermal += 2;
44   robotics += 1;
45 }
46
47 // Career Goal Analysis
48 if(user.goal === "EV Industry"){
49   thermal += 3;
50 }
51
52 if(user.goal === "Research"){
```

</> Execute Code

"User Interest Analysis for Career Path Recommendations"

🔗 Generate your action with AI

Ask Botpress AI to edit code



```
49 | thermal += 3;
50 | }
51 |
52 | if(user.goal === "Research"){
53 |   thermal += 2;
54 |   robotics += 2;
55 | }
56 |
57 | if(user.goal === "Robotics & Automation"){
58 |   robotics += 4;
59 | }
60 |
61 | // Save scores
62 | workflow.thermalScore = thermal;
63 | workflow.roboticsScore = robotics;
64 | workflow.manufacturingScore = manufacturing;
65 | // Find highest score
66 | let maxScore = Math.max(thermal, robotics, manufacturing);
67 |
68 | if(maxScore === thermal){
69 |   workflow.bestDomain = "Thermal and Fluid Sciences";
70 | }
71 |
72 | else if(maxScore === robotics){
73 |   workflow.bestDomain = "Design and Robotics";
74 | }
75 |
76 | else{
```

</> Execute Code

"User Interest Analysis for Career Path Recommendations"

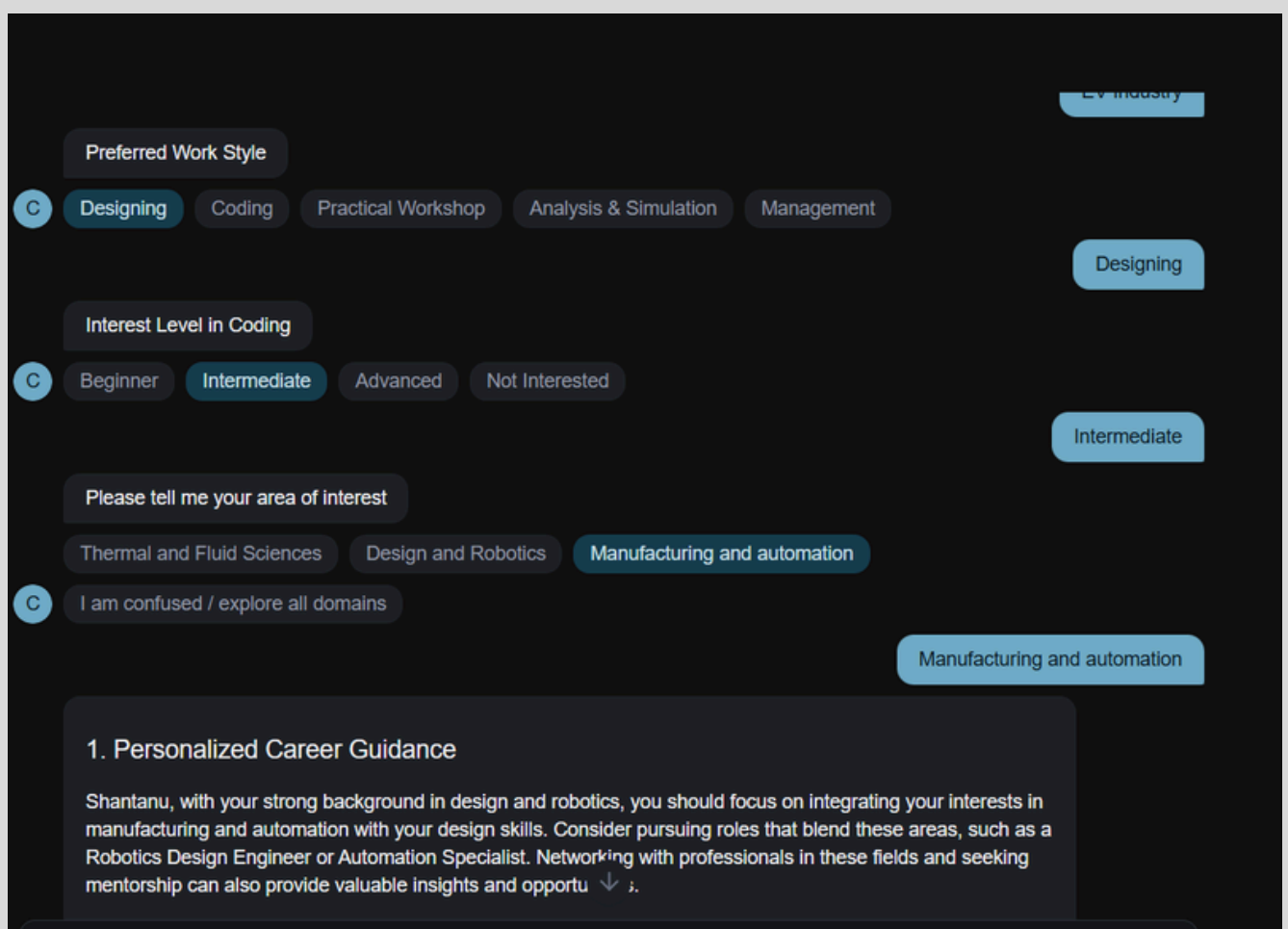
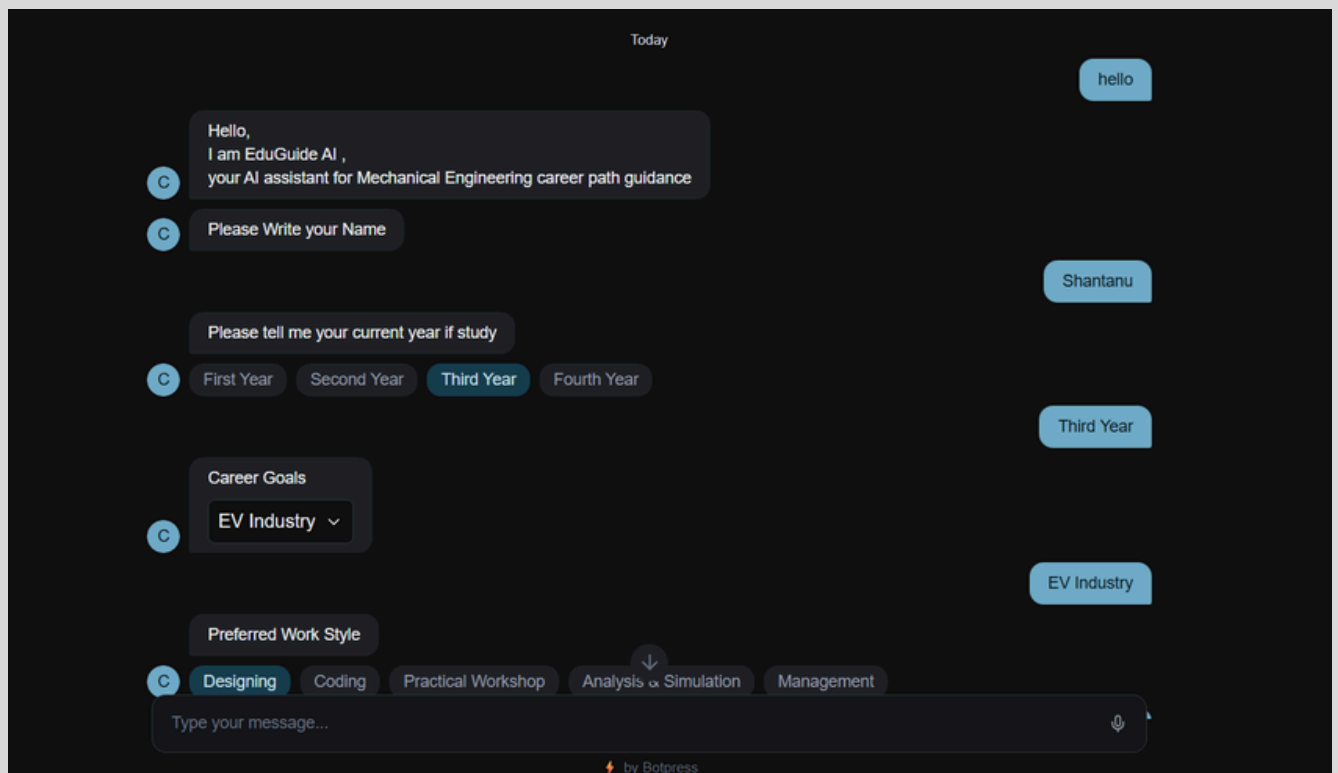
🔗 Generate your action with AI

Ask Botpress AI to edit code



```
76 | else{
77 |   workflow.bestDomain = "Manufacturing and Automation";
78 | }
79 |
80 | // Student Profile Type
81 | if(workflow.bestDomain === "Design and Robotics" && user.coding !== "Not Interested"){
82 |   workflow.studentType = "Automation-Oriented Student";
83 | }
84 | else if(workflow.bestDomain === "Thermal and Fluid Sciences"){
85 |   workflow.studentType = "Energy Systems Enthusiast";
86 | }
87 | else{
88 |   workflow.studentType = "Manufacturing & Production Specialist";
89 | }
90 | // student year type
91 | if(user.year === "First Year"){
92 |   workflow.focus = "Explore fundamentals and discover interests";
93 | }
94 | else if(user.year === "Second Year"){
95 |   workflow.focus = "Build technical skills and projects";
96 | }
97 | else if(user.year === "Third Year"){
98 |   workflow.focus = "Specialize and prepare for internships";
99 | }
100 | else{
101 |   workflow.focus = "Placement preparation and career specialization";
102 | }
```

Profile analyser logic



1. Personalized Career Guidance

Shantanu, with your strong background in design and robotics, you should focus on integrating your interests in manufacturing and automation with your design skills. Consider pursuing roles that blend these areas, such as a Robotics Design Engineer or Automation Specialist. Networking with professionals in these fields and seeking mentorship can also provide valuable insights and opportunities.

2. Recommended Electives

- **Robotics Systems Design:** Deepen your understanding of robotics components and systems.
- **Automation and Control Systems:** Learn about automated systems and their applications in manufacturing.
- **3D Modeling and CAD:** Enhance your design skills with advanced modeling software.
- **Machine Learning for Robotics:** Explore how AI can be integrated into robotic systems.
- **Human-Robot Interaction:** Understand the design aspects of user-friendly robotic systems.

3. Software/Tools to Learn

- **SolidWorks or AutoCAD:** For 3D modeling and design.
- **MATLAB:** For simulations and algorithm development in robotics.
- **ROS (Robot Operating System):** Essential for programming and controlling robots.
- **Fusion 360:** For integrated CAD, CAM, and CAE.
- **Arduino or Raspberry Pi:** For hands-on projects in robotics and automation.

4. Mini Projects



- **Fusion 360:** For integrated CAD, CAM, and CAE.
- **Arduino or Raspberry Pi:** For hands-on projects in robotics and automation.

4. Mini Projects

- **Automated Plant Watering System:** Design a system using sensors and microcontrollers.
- **Robotic Arm:** Build and program a simple robotic arm for specific tasks.
- **Smart Home Automation:** Create a prototype that controls home appliances via a mobile app.
- **Line Following Robot:** Develop a robot that can follow a line using sensors.
- **3D Printed Drone:** Design and print a drone frame and integrate basic flight controls.

5. Internship Suggestions

- **Robotics Companies:** Look for internships at companies specializing in robotics design and automation.
- **Manufacturing Firms:** Seek positions that focus on automation and process improvement.
- **Research Labs:** Consider internships in academic or industrial research labs focusing on robotics and automation.
- **Startups:** Explore opportunities in tech startups that focus on innovative automation solutions.

6. Career Opportunities

- **Robotics Engineer:** Design and develop robotic systems.
- **Automation Engineer:** Focus on automating manufacturing processes.
- **Product Designer:** Work on the design of robotic products.
- **Systems Engineer:** Integrate various components of robotic systems.
- **Research Scientist:** Engage in advanced research in robotics and automation technologies.

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- **Research Scientist:** Engage in advanced research in robotics and automation technologies.

7. 6-Month Roadmap

Month 1-2: Skill Development

- Enroll in online courses for CAD software and ROS.
- Start a mini project (e.g., Automated Plant Watering System).

Month 3: Networking

- Attend workshops and seminars related to robotics and automation.
- Join relevant online forums and LinkedIn groups.

Month 4: Electives and Projects

- Register for recommended electives.
- Continue working on mini projects, focusing on complexity.

Month 5: Internship Applications

- Update your resume and portfolio showcasing your projects.
- Apply for internships in robotics and automation.

Month 6: Interview Preparation

- Prepare for technical interviews by practicing coding and design problems.

- Continue working on mini projects, focusing on complexity.

Month 5: Internship Applications

- Update your resume and portfolio showcasing your projects.
- Apply for internships in robotics and automation.

Month 6: Interview Preparation

- Prepare for technical interviews by practicing coding and design problems.
- Conduct mock interviews with peers or mentors.

By following this structured approach, you can effectively prepare for a successful career in design and robotics while aligning with your interests in manufacturing and automation.

Want to go back to previous menu ?

To start

Area of interest

No

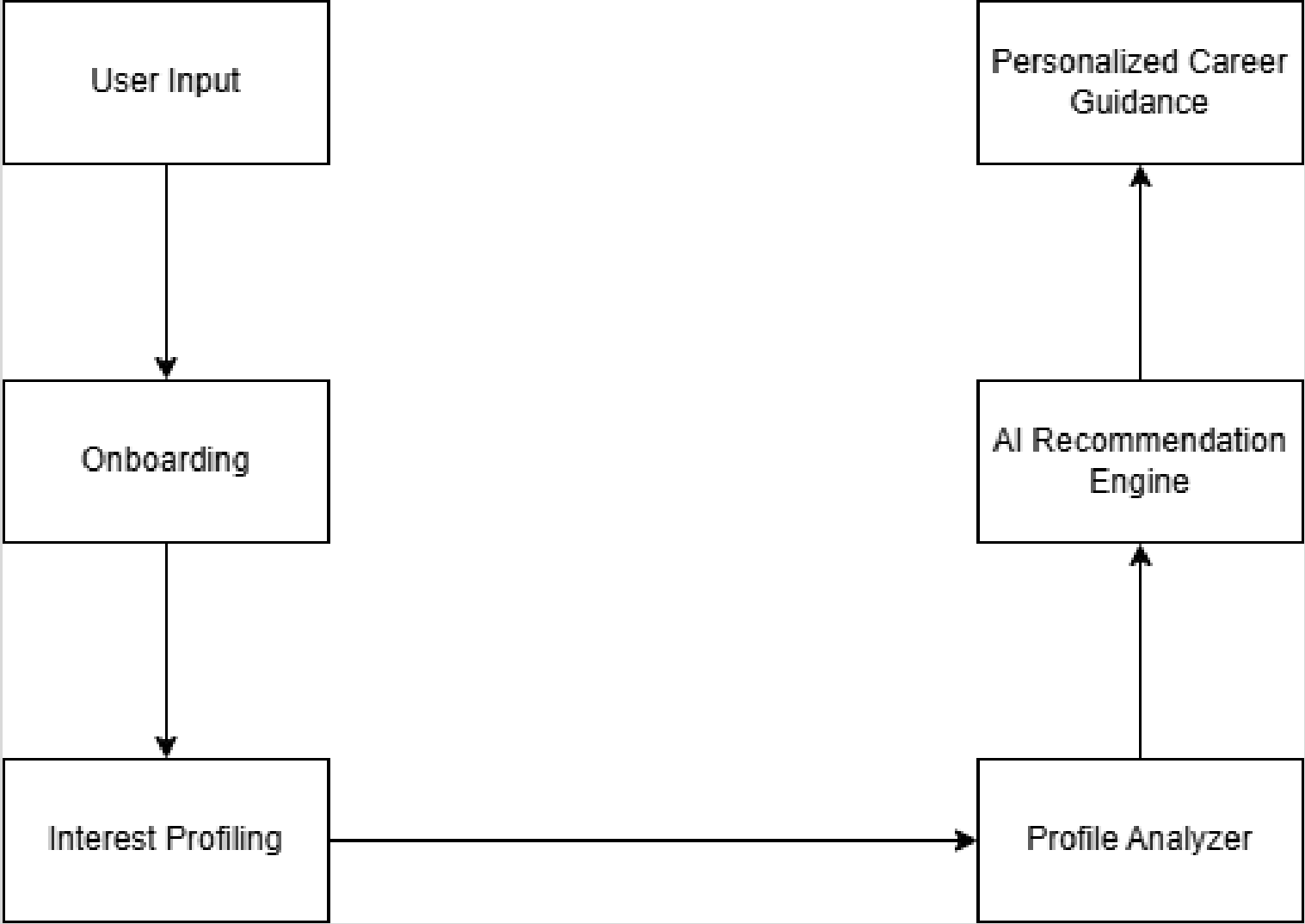
No

Delivered

I hope this helps you navigate your academic journey! Feel free to ask me any other specific questions about these fields.

Chatbot output

Workflow Architecture



Future Scope

The EduGuide AI project has a lot of potential to develop and improve in the future. The platform can be made more intelligent, interactive, and industry-focused by integrating a number of innovative AI capabilities and student support features, even if the current system concentrates on offering Mechanical Engineering students individualized career coaching.

- **Voice-Based Interaction:** By adding speech recognition and voice response systems, students can use voice commands to communicate with the chatbot, increasing the platform's usability and accessibility.
- **Support for Mobile Applications:** Creating a specialized mobile application for iOS and Android would increase accessibility and allow students to get career advice at any time and from any location.
- **Analysis of Resumes and Suggestions for Improvement:** AI-powered resume evaluation can help students in identifying areas where their resumes fall short and offer recommendations for enhancing formatting, technical proficiency, project descriptions, and overall appearance.
- **System for Job and Internship Recommendations:** Based on a student's interests, abilities, academic standing, and desired career field, the chatbot can be linked with internship and employment portals to provide opportunities.

- **Analysis of Industry Trends in Real Time:** The chatbot may offer up-to-date information about new technologies, in-demand skills, and potential career paths in mechanical engineering by incorporating real-time industry data and market trends.
- **AI-Powered Analysis of Skill Gaps:** In order to identify skills that are lacking and suggest learning routes for growth, the chatbot can evaluate a student's present skill set and compare it with industry requirements.
- **Assist in Placement Interview Preparation:** To assist students in effectively prepare for placements, future versions might include HR interview preparation modules, technical question banks, aptitude practice, and mock interview simulations.

Additionally, the chatbot can be expanded beyond Mechanical Engineering and adapted for students from other engineering branches such as Computer Engineering, Civil Engineering, Electrical Engineering, Electronics Engineering, and Information Technology. This would make the platform a scalable AI-driven educational guidance system capable of supporting students across multiple disciplines.